

Variable description table

Variable name	Data type/variable type	Purpose
h1, h2	int, int	to store and print hours.
m1, m2	int, int	to store and print minutes.
hour	int	to store added hours.
minute	int	to store added minutes.

Unsolved Problems

1. Make a class **Rectangle** to find the area of a rectangle using : **area = length x breadth**

The **instance variables** of class are **length** and **breadth**. Its member functions are:

- Rectangle()** - default constructor to assign 0 to **length** and **breadth**.
- Rectangle(float L, float B)** - to store value of **L** to **length** and **B** to **breadth**.
- float FindArea()** - to calculate the area of rectangle and return.

Write a **main()** function to create an object with parameters and invoke the above functions print the area.

2. Make a class **FibSeries** to generate Fibonacci series: 0, 1, 1, 2, 3, 5, 8, 13, 21,

(The Fibonacci series is generated by adding two previous terms by taking 0, 1 as starting terms.)

Example : the starting values are 0, 1 then if add $(0+1)=1$ (this is 3rd term), then add $(1+1)=2$ (this is 4th term and so on...)

Private variables of the class are **first, second, third** (integer types).

Public functions of the class are:

- FibSeries()** - to store 0 to **first** and 1 to **second** (as two starting terms).
- void PrintSeries()** - to print Fibonacci series up to 30 terms using **while loop**.

Write a **main()** function to print the Fibonacci series by calling suitable functions.

3. Create a class **factorial** whose members are as given:

Private data members of the class : num, f (integer variables)

Public member functions of the class :

- factorial()** - a default constructor to assign 1 to 'f'.
- factorial(int n)** - a parameterized constructor to assign **num = n**.
- int GetFactorial()** - to find and return the factorial of **num**.

Write a **main()** function to input a number and by invoking methods, print the factorial.

4. A class **finder** is created with following members:

Variables of the class : Start, Last (integer variables as start and last limits.)

Member functions of the class :

- finder(int s, int L)** - a parameterized constructor to assign data '**Start = s**' and '**Last = L**'.
- int checkPrime(int d)** - to find and return 1 if 'd' is prime otherwise return -1.
- void printPrime()** - to print only prime numbers between **start** to **last** limits (both inclusive) by invoking function **checkPrime()**.

Write a **main()** function to input **start** and **last** limits and print prime numbers by calling suitable functions.

5. A class **SUM** is defined as follows:

Data members : num (long integer variable)

Member functions :

(i) **SUM(long n)**- parameterized constructor to store value of variable **num = n**.

(ii) **long FindSum()**- to find and return sum of all the digits present in **num**.

Write a **main()** program to input a number, create an object and print the sum of the digits by invoking suitable function along with the input number.

6. A class **Reverse** is defined as follows:

Data members : num (long integer variable)

Public member functions :

(i) **Reverse(long n)**- parameterized constructor to store value of variable '**num = n**'.

(ii) **long findReverse()**- to find and return reverse of the number present in **num**.

Write a **main()** function to input a number, create a suitable object and by invoking the function print the reverse of the number along with the input number.

7. Design a class **Table** as follows:

Data members : num (an integer member)

Member functions/methods :

(i) **Table(int v)**- a constructor to assign '**v**' to **num**.

(ii) **void print()**- print the mathematical table of **num** from 1 up to 10 in the following format :

Let num = 4

Output :

4 x 1 = 4

4 x 2 = 8

Write a **main()** function to create an object and print the table by invoking above functions.

8. The sum of two distances is calculated as :

Distance 1 = 10 feet 24 inches

Distance 2 = 5 feet 16 inches

Sum of distances = 18 feet 4 inches

(where 12 inches = 1 feet)

A class **Distance** has following members:

Instance variable/data members : f1, f2 (integers to store two feet values).

in1, in2 (integers to store two inches).

Member functions/methods :

(i) **public Distance(int f, int n1, int ff, int n2)**- constructor to initialize '**f**' to '**f1**', '**n1**' to '**in1**', '**ff**' to '**f2**' and '**n2**' to '**in2**'.

(ii) **public void showdistance()**- to display the both the distances with suitable messages.

(iii) **public void SumOfDistance()**- to find sum of distances using the above method and print the sum of distances.

Write a **main()** function to input the two distances and print entered and sum of distances by invoking suitable functions.

[Hint : Refer solved problem 4]

9. Design a class **Account** for a brief bank status. Include following members of class :

Data members :

Account number

Type of account (saving or current)

Balance amount in the account

Member functions of the class :

- (i) Constructor to assign values to data members of class.
- (ii) To deposit an amount.
- (iii) To withdraw an amount after checking the balance (minimum balance should be Rs. 1000/-)
- (iv) To display the account number, type of account and balance amount.

Design a **main()** function to create the object and print the required data by invoking above methods.

A class **Compound** is created to calculate the compound interest using :

$$CI = [p \cdot (1 + r/100)^t] - p$$

Where **p** – is principal amount, **r** – rate of interest and **t** – time period.

Data members of the class : Pamt, rate (double data type to store principal amount, rate of interest)
Time (integer to store time period)

Member functions of the class :

- (i) **Compound(double p, double r, double t)** – constructor to assign '**Pamt = p**', '**Rate = r**' and '**Time = t**'.
- (ii) **double FindInterest()** – to find and return compound interest.
- (iii) **void printData()** – to print principal amount, rate of interest and time period.

Write a **main()** function to input required data, create an object and by invoking suitable functions print the input data and the compound interest.

Design a class **Product** with following members:

Data members : num (long integer data)

Member functions :

- (i) **Product()** – a default constructor to assign a number of your choice to variable **num**.
 - (ii) **long ProductOfDigits()** – to print the number and return product of all the digits present in **num**.
- Write a **main()** method to print the required output by invoking suitable methods.

Design a class **roots** with following members:

Data members : a, b, c, X1, X2 (all float data members)

Member functions :

- (i) **roots(float x, float y, float z)** – a parameterized constructor to assign '**a = x**', '**b = y**', '**c = z**'.
- (ii) **void FindRoots()** – to find and print the roots **X1, X2** from the following relation:

$$X1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$X2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Write a **main()** program to create object of class and print the roots **X1, X2** using above methods.

13. Define a class **student** described as below :

Data members /instance variables :

roll_number, age, m1, m2, m3 (marks in 3 subjects) maximum, average

Member Methods :

- (i) A parameterized constructor to initialize data members.
- (ii) To accept the details of a student.
- (iii) To compute the average and maximum out of three marks.
- (iv) To display the name, age, marks in 3 subjects, maximum and average.

Write a **main()** method to create an object and call above member methods.

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